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PROFIT AND LOSS
DISCOUNT

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Introduction

Profit, Loss, & Discount covers core concepts like **Cost Price (CP)**, **Selling Price (SP)**, **Profit/Loss**, **Marked Price (MP)**, and **Discount**, focusing on calculating percentages, finding missing values (CP, SP, MP), and solving problems involving successive discounts, false weights (dishonest shopkeepers), and cash discounts, using formulas like Profit = SP - CP and % Profit = (Profit/CP) x 100.

Concept and Definitions

Cost Price(CP) : Price at which an item is bought/produced.

Selling Price(SP) : Price at which an item is bought/produced.

Profit(P) : SP > CP (Profit = SP - CP).

Loss(L): CP > SP (Loss = CP - SP).

Marked Price(MP) : Labeled/Listed Price before discount.

Discount : Reduction from MP (Discount = MP – SP).

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Key Formulae

Profit %: (Profit / CP) × 100.

Loss %: (Loss / CP) × 100.

Discount %: (Discount / MP) × 100.

Finding SP:

With Profit %: SP = CP (1 + Profit%/100).

With Loss %: SP = CP (1 - Loss%/100).

Finding CP:

With SP & Profit %: CP = SP / (1 + Profit%/100).

With SP & Loss %: CP = SP / (1 - Loss%/100).

Successive Discounts: Calculating equivalent single discount for multiple discounts (e.g., m% and n%).

Questions

Ques.1 A retailer increases the price of a product by 30%. If the new price is ₹1,040, Find the original cost price.

a) 800

b) 750

c) 850

d) 940

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Questions

Ques.2 A trader buys goods for ₹5,000 and marks them at ₹7,000. He offers a discount of 20% and sells them. What is his profit percentage?

a) 12%

b) 20%

c) 10%

d) 15%

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Questions

Ques.3 If selling price of 20 articles equals the cost price of 25 articles, what is the profit percentage?

a) 20%

b) 25%

c) 30%

d) 35%

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Questions

Ques.4 A product is marked at ₹2,400. A shopkeeper gives a discount of 10% on the marked price and earns a profit of 8%. What is the cost price?

a) Rs. 2000

b) Rs. 2050

c) Rs. 2100

d) Rs. 2150

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Questions

Ques.5 A merchant uses a 1 kg weight that is 100 grams less. He marks up his goods by 25% and offers a discount of 10%. What is his overall profit percentage?

a) 16%

b) 20%

c) 25%

d) 18%

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Questions

Ques.6 By selling an article at ₹900, a shopkeeper incurs a loss of 10%. At what price should he sell it to gain 15%?

a) Rs. 1150

b) Rs. 1200

c) Rs. 1180

d) Rs. 1250

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Questions

Ques.7 A man buys an article for ₹2,000 and sells it at 10% profit. If he had bought it at a 10% lower price and sold it at a price ₹100 higher, what would have been his profit percentage?

a) 25%

b) 20%

c) 15%

d) 27.78%

Questions

Ques.8 A man sells an article at 20% profit. If he had sold it for ₹40 more, his profit percentage would have been 30%. What is the cost price?

a) Rs.400

b) Rs.360

c) Rs.320

d) Rs.300

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Questions

Ques.9 A trader bought 1,000 articles at ₹5 each. He sold 20% of the articles at a 20% profit, 50% of the articles at a 10% profit, and the rest at a 5% loss. What is his overall profit or loss percentage?

a) 8% Profit

b) 8% Loss

c) 6% Profit

d) 6% Loss

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Questions

Ques.10 The marked price of an article is ₹1,200, and the shopkeeper gives a discount of ₹200 while still earning a profit of ₹100. What is the cost price?

a) Rs. 900

b) Rs. 1000

c) Rs. 850

d) Rs. 800

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Questions

Ques.11 A retailer buys an article at ₹600 and marks it 50% above the cost price. He gives two successive discounts of 20% and 10%. What is his final selling price? **[Infosys 2025]**

a) Rs. 900

b) Rs. 648

c) Rs. 855

d) Rs. 899

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Questions

Ques.12 A man buys an article for ₹2,000 and sells it to his friend at a profit of 25%. His friend sells it to another person at a profit of 10%. What is the final selling price of the article?

a) Rs. 2500
b) Rs. 2750
c) Rs. 3000
d) Rs. 3100

Questions

QUES.13 A shopkeeper reduces the cost price of an article by ₹50 and still makes a profit of 20% by selling it at ₹600. What was the original cost price?

a) Rs. 450
b) Rs. 500
c) Rs. 550
d) Rs. 520

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Questions

QUES.14 A shopkeeper sells two articles at ₹4,000 each. On one, he incurs a loss of 20%, and on the other, he gains 25%. What is his overall gain or loss percentage?

a) 1.08% Gain
b) 2.8 % Gain
c) 2.44% Loss
d) 1.08% Loss

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Questions

QUE.15 A trader purchases 100 identical items at a total cost of ₹10,000. He marks each item at 50% above its cost price. Due to defects, 20 items are unsellable and discarded. The remaining items are sold at a discount of 20% on the marked price. What is the overall profit or loss percentage for the trader?

a) 10% Profit
b) 4% Loss
c) 12% Profit
d) 8% Loss

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Thank You